

TS-4 and TS-4e Firmware Release Process

Document ID: eng-ts4-firmware-release-process. Owner: Tessera Firmware Team (release manager role rotates per release). Last updated: 9 June 2025.

Scope

This process covers firmware for the Tessera TS-4 gateway and the TS-4e edge node, from release candidate to general availability through TesseraCloud over-the-air (OTA) updates.

Release types

Type | Version change | Example | Typical cadence

Major | First number | 4.x to 5.0 | Yearly

Minor | Second number | 5.2 to 5.3 | Quarterly

Patch | Third number | 5.2.0 to 5.2.1 | As needed

Hotfix | Patch released outside the normal cadence | 5.2.2 for a security fix | As needed

Stages

1. Release candidate built in Forge by a release pipeline on the release-tagged runners, including the hil-test stage on the tessera-tagged Penang runners and the sign stage.
2. The release manager declares a firmware release freeze on the release branch. During the freeze, Forge runner token rotation is also paused on the Penang runners.
3. Staging soak: the candidate runs on the 40 test TS-4 gateways and 25 TS-4e edge nodes connected to TesseraCloud staging for at least 7 days (major and minor) or 48 hours (patch).
4. Pilot ring: OTA to 2% of production devices that belong to customers enrolled in the early-access programme, for 7 days.
5. Staged rollout: 10%, then 25%, then 50%, then 100% of the fleet, with at least 48 hours between steps.
6. General availability announcement to customers.

Rollout halt criteria

The rollout is halted automatically if, in any ring:

- more than 0.5% of updated devices fail to reconnect within 30 minutes, or
- more than 1% of updated devices report a watchdog reset within 24 hours, or
- any SEV2 or higher incident is linked to the release.

Halted rollouts can only be resumed by the release manager with approval from the Head of Platform Engineering.

Reconnect behaviour requirement

Since gateway firmware 5.0.0 (September 2024) the TS-4 gateway uses jittered reconnect backoff and paced buffer replay, an action item from INC-2024-031. Any release that changes reconnect or replay behaviour must pass a reconnect-storm test in staging (all 40 test gateways reconnecting at once) and requires approval from the Head of Platform Engineering.

Minimum supported versions

Device | Minimum supported firmware | End of support for older versions

TS-4 gateway | 5.2.0 | TesseraCloud requires 5.2.0 or later from 1 October 2025; older gateways show as Degraded

TS-4e edge node | 2.6.1 | Node 2.6.1 is deprecated on gateway 5.3.0; see the product firmware

compatibility matrix

Records

The release manager records every release in the release register with the version, Forge pipeline ID, signing timestamp, soak results and the rollout ring dates.